

RESEARCH ARTICLE

The impact of energy consumption on environmental pollution in newly industrialized countries

Ökkeş ÇELEN ^{1*}, Aziz Can ŞENSAZLI ²

¹ Manisa Celal Bayar University, Faculty of Business, Manisa, Türkiye

² Manisa Celal Bayar University, Salihli Vocational School, Manisa, Türkiye

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ABSTRACT

This study examines the impact of energy consumption on the environment for ten newly industrialized countries through panel data methods using the data covering the period 1970-2022. Due to the presence of cross-section dependence in the panel data, the CIPS unit root test is performed. The cointegration test shows the existence of a long-run relationship between energy consumption and ecological footprint. Long-run heterogeneous panel elasticities are estimated through the Common Correlated Effects Mean Group (CCEMG) estimator and the Augmented Mean Group (AMG) estimator. Empirical results reveal that energy consumption increases environmental pollution in all examined countries, with statistically significant effects observed for Brazil, China, India, Indonesia, Malaysia, Mexico, the Philippines, South Africa, Thailand, and Türkiye. Causality results show the presence of unidirectional causality from energy consumption to ecological footprint. This empirical analysis provides policy-makers with policy recommendations for the effective and efficient use of renewable energy resources—such as providing incentives and promoting environment-related technologies—for a sustainable environment in the long run.

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INTRODUCTION

In recent years, with climate change and global warming, environmental sustainability has become a topic of discussion [1]. Increasing industrialization and urbanization due to the globalization process have made an impact on growth rates [2]. As a result of this situation, there has been an increase in energy demand with the increase in growth rates. As energy demand is met with non-renewable energy resources, the negative impacts of globalization and growth on the environment have started to emerge.

These developments have prompted a growing demand for alternative indicators that account for broader environmental degradation beyond merely air pollution. In this context, the ecological footprint emerges as a more comprehensive metric, as it encompasses not only carbon emissions but also environmental consequences such as land use, natural resource consumption, and waste generation. Unlike traditional indicators such as CO₂ emissions, the ecological footprint reflects the total pressure that human activities exert on the planet's biological carrying capacity. Therefore, the adoption of this indicator enables a more accurate assessment of the environmental impacts of energy consumption and economic growth, particularly in countries

undergoing rapid industrialization and structural transformation.

Studies in the literature use CO₂ emissions to examine the impact of growth or energy consumption on the environment [3]. However, using data such as CO₂ or greenhouse gas emissions to analyze the impact of energy consumption on the environment only reveals the effect of energy consumption on air pollution. This represents a shortcoming in terms of determining appropriate policies; in that case, a more appropriate variable should be used to investigate the impact of energy consumption on the environment. This study contributes to the existing literature in the following ways: examining the impact of energy consumption on the environment through panel time series analysis using a large dataset for ten newly industrialized countries; using ecological footprint data to measure the impact of energy consumption on the environment; checking the stationarity of the series with the Pesaran [4] CIPS (Cross-sectionally Augmented Im, Pesaran, and Shin) test, which takes into account cross-sectional dependence; using the Westerlund [5] cointegration test, one of the second-generation cointegration tests, to assess the existence of a long-run relationship between the series; utilizing the Eberhardt and Teal [6] AMG (Augmented Mean Group) estimator and Pesaran [7] CCE (Common Correlated Effects) estimator,

* Corresponding Author: okkes.celen@cbu.edu.tr (Ökkeş Çelen)



which are panel error correction models that can be used when there is cross-sectional dependence in the series and the panel is heterogeneous; examining the causality between the variables with the Dumitrescu and Hurlin [8] Granger causality test, which is a heterogeneous panel causality test.

The sections in the remaining part of this study are as follows: Section II presents the related literature, Section III presents the methodology used, Section IV presents the dataset and analysis results, and Section V presents the conclusion.

LITERATURE REVIEW

There are numerous studies in the literature on testing the Environmental Kuznets curve and on energy consumption and its impacts on the environment (Table 1). In studies testing the EKC hypothesis, Beyene and Kotosz [9] found no significant association between economic activities and CO₂ emissions in 12 East African countries. Sarkodie and Owusu [10], analyzing Senegal, reported that energy consumption, financial development, and industrialization contributed to rising carbon emissions, while urbanization and GDP had a mitigating effect in the long run. Similarly, Azam et al. [11] confirmed a positive relationship between CO₂ emissions and economic growth. In their study on Pakistan, Mohiuddin et al. [12] identified a long-run unidirectional causal relationship from energy consumption to CO₂ emissions. In Sri Lanka, Sarkodie and Owusu [13] reported a long-run equilibrium between energy consumption and industrialization, with bidirectional causality. They also observed a unidirectional causality from CO₂ emissions to energy use.

Wang et al. [3] concluded that shocks in CO₂ emissions exerted limited feedback effects on GDP and energy consumption. Apergis and Öztürk [14] confirmed the EKC hypothesis in Asian countries, while Osabuohien et al. [15] validated it for a group of African nations. Baek [16], in a study covering 12 nuclear energy-intensive countries, did not find support for the EKC hypothesis but noted that nuclear energy contributed to long-run reductions in CO₂ emissions.

Remuzgo and Sarabia [17] found that global inequality in CO₂ emissions declined by 22% between 1990 and 2010, driven by global development. Lin et al. [18], in a study on Nigeria, reported no significant impact of industrialization on CO₂ emissions. In Malaysia, Gul et al. [19] identified a unidirectional causality from energy consumption to CO₂ emissions. Salahuddin et al. [20], focusing on the Gulf Cooperation Council (GCC) countries, observed a unidirectional causality from electricity consumption to CO₂ emissions. Using the ARDL approach, Saidi and Mbarek [21] tested the EKC hypothesis in 19 developing countries between 1990 and 2013 and found no evidence supporting the EKC for this group.

Tiwari et al. [22] investigated the EKC hypothesis for India and concluded that it holds in both the short and long run. Similarly, Shahbaz et al. [23] confirmed the long-run validity of the EKC in Pakistan. For the BRICS countries, Pao and Tsai [24] found support for the EKC hypothesis and observed a bidirectional causality between CO₂ emissions and foreign direct investment. In their study on Pakistan, Nasir and Rehman [25] identified positive effects of energy consumption and foreign trade on CO₂ emissions and confirmed the presence of the EKC. Acaravci and Ozturk [26], in their analysis of European countries, established a long-run equilibrium relationship between energy use and CO₂ emissions driven by economic growth and validated the EKC

in Denmark and Italy. In the case of China, Jalil and Mahmud [27] also reported results consistent with the EKC hypothesis.

Only a few studies treat energy consumption as the dependent variable, and these generally focus on its relationship with economic growth. Lee and Chang [28], examining 16 Asian countries between 1971 and 2002, found a long-run positive relationship between energy consumption and real GDP, along with unidirectional causality running from energy consumption to growth. In contrast, short-run causality was not observed. Romano and Scandurra [29] studied Italian regions between 1980 and 2007 using panel cointegration and vector error correction models. Their findings revealed long-run cointegration and bidirectional causality in both the short and long term.

Sarkodie and Strezov [30], analyzing China, India, Iran, Indonesia, and South Africa between 1986 and 2016, reported that energy consumption strongly and positively influenced greenhouse gas emissions. Using the ARDL approach for India, Mahalik and Mallick [31] concluded that energy consumption was positively influenced by urban population and negatively influenced by financial development, economic growth, and industrial output. Shahbaz et al. [32] found that globalization and financial development reduced energy consumption, while economic growth and urbanization contributed to increased energy demand. Bashir et al. [33], studying Indonesia, discovered a short-run causal relationship between CO₂ emissions and energy consumption.

Despite the growing relevance of broader sustainability indicators, studies employing the ecological footprint remain scarce. One such study by Destek and Sarkodie [1] found a bidirectional causal relationship between the ecological footprint and economic growth.

Recent studies increasingly explore the complex relationships between economic growth, energy consumption, and environmental degradation through both traditional and alternative environmental indicators. In this context, Recepoglu et al. (41) investigated the impact of ecological footprint components on agricultural productivity in MENA countries over the 1992–2021 period. Their findings revealed that all components negatively affected productivity, with forest footprint exerting the strongest adverse effect. Similarly, Hayaloglu et al. (42) approached the EKC hypothesis from an institutional quality perspective and found that institutional structures significantly increased environmental degradation up to a certain threshold, after which the effect reversed in SADC countries. Qing et al. (43) examined the relationships among green energy consumption, environmental degradation, economic development, trade openness, and oil prices, concluding that environmental degradation reduces green energy consumption, whereas oil prices and economic growth increase it. Kostakis (44), focusing on ASEAN countries, found that the EKC hypothesis held only for Singapore, and that renewable energy supports environmental sustainability, while financial openness contributes to rising CO₂ emissions. Bashir et al. (45) analyzed the effects of energy transition, natural resource consumption, geopolitical risks, and banking development on environmental sustainability. Their empirical results showed that while energy transition and banking developments contribute positively to sustainability, natural resource use, economic growth, and geopolitical risk exacerbate environmental degradation.

Table 1. Summary of literature review

Reference	Period	Field of Study	Variables	Method	Results
Destek and Sarkodie [1]	1977-2013	11 newly industrialized countries	Ecological Footprint (EF), Financial Development (FD), Energy Consumption (EC), and GDP	AMG (Augmented Mean Group) estimator and heterogeneous panel causality method	It was concluded that there was bidirectional causality between economic growth and ecological footprint.
Sarkodie and Owusu [10]	1980-2011	Senegal	Carbon dioxide emissions (CO ₂), Electricity Consumption, Financial Development, Industrialization (IND), Urbanization (URB), and GDP	NIPALS (Nonlinear iterative partial least squares) regression	They concluded that EC, FD, and IND increased Co ₂ , while URB and GDP reduced CO ₂ in the long run.
Azam et al. [11]	1971-2013	USA, China, India, and Japan	Carbon dioxide, Trade Openness (TO), Energy Consumption	FMOLS (Fully Modified Ordinary Least Squares) method	A positive relationship between CO ₂ and GDP was identified in the USA, China, and Japan.
Wang et al. [3]	1990-2012	China	Carbon dioxide, Energy Consumption, and GDP	VECM (Vector Error Correction Model)	Shocks in CO ₂ emissions had a slight impact on EC and GDP.
Mohiuddin et al. [12]	1971-2013	Pakistan	Carbon dioxide, Energy Intensity (EI), Population (P), and GDP	VECM	There was a long-run relationship and unidirectional causality from EC to carbon dioxide emissions.
Sarkodie & Owusu [13]	1971-2012	Sri Lanka	Carbon dioxide, Energy Consumption, Financial Development, Industrialization, and GDP	ARDL (Autoregressive Distributed Lag) analysis and Neural Network	A long-run bidirectional causality relationship was determined.
Apergis and Ozturk [14]	1990-2011	Asian countries	CO ₂ emissions, Land (LM), Population density (P), Industry shares (IND), and GDP per capita	GMM (Generalized Method of Moments)	It was determined that the EKC (Environmental Kuznets Curve) hypothesis was valid.
Baek [16]	1980-2009	12 countries	CO ₂ emissions, Energy Consumption, Nuclear Energy (NUE), and GDP	DOLS (Dynamic Ordinary Least Squares) and FMOLS	The validity of the EKC hypothesis could not be established; the presence of nuclear energy was found to reduce CO ₂ emissions.
Remuzgo and Sarabia [17]	1990-2010	World data	CO ₂ emissions, Electricity Intensity, Population, and GDP	Decomposition	Global inequality in CO ₂ emissions declined by 22% from 1990 to 2010.
Lin et al. [18]	1908-2011	Nigeria	CO ₂ emissions, Energy Intensity, Population, Industrialization, and GDP	VECM	An inverse relationship was detected between IND (Industrialization) and Co ₂ emissions.
Gul et al. [19]	1975-2013	Malaysia	CO ₂ emissions, Energy Consumption	MEBM (Maximum Entropy Bootstrap - MEboot)	In the EKC hypothesis test, a unidirectional causality from EC to CO ₂ emission was detected.
Salahuddin et al. [20]	1980-2012	GCC member countries	Carbon dioxide emissions, Electricity Consumption, Financial Development, and GDP	DOLS, FMOLS, and DFE (Dynamic Fixed Effect) Model	There was no significant short-term relationship; a positive long-run relationship was detected between Co ₂ emissions and EC.
Saidi and Mbarek [21]	1990-2013	19 emerging economies	CO ₂ emissions, Income (INC), Trade Openness, Financial Development, and Urbanization	ARDL	The existence of the EKC hypothesis could not be determined.
Osabuohien et al. [15]	1995-2010	African countries	CO ₂ Emissions, Electricity Consumption, Population, and GDP	Panel DOLS	The existence of the EKC hypothesis could be determined.
Tiwari et al. [22]	1966-2011	India	CO ₂ emissions, Coal Consumption (CC), Trade Openness, and GDP	ARDL	The presence of the EKC hypothesis could be determined.
Romano and Scandurra [29]	1980-2007	Regions of Italy	Energy Consumption, Real Gross Domestic Product (real GDP)	Panel vector error correction models and panel cointegration analysis	A bidirectional causality relationship was found in the short run and long run.
Shahbaz et al. [23]	1971-2009	Pakistan	CO ₂ emissions, Energy Consumption, Trade Openness, and GDP	ARDL	The existence of the EKC hypothesis could be determined.
Pao and Tsai [24]	1992-2007	BRIC countries	CO ₂ emissions, Energy Consumption, Foreign Direct Investment (FDI), and GDP	VECM and Granger Causality Test	The existence of the EKC hypothesis and bidirectional causality between FDI and CO ₂ emissions were detected.

Nasir and Rehman [25]	1972-2008	Pakistan	CO2 emissions, Foreign Trade (FT), Energy Consumption, and Income	VECM	The presence of EKC and the positive effects of FT and EC on CO2 emissions were detected.
Acaravci and Ozturk [26]	1960-2005	European countries	CO2 emissions, Energy Consumption, and GDP	ARDL	Long-run relationships and the presence of EKC were detected in Denmark and Italy.
Lee and Chang [28]	1971-2002	16 Asian Countries	Energy Consumption, Real GDP	Panel unit root, heterogeneous panel cointegration and panel-based error correction models	They found a long-run unidirectional causality relationship running from energy consumption to economic growth.
Jalil and F. Mahmud [27]	1992-2007	China	CO2 emissions, Energy Consumption, Foreign Trade and GDP	VECM and Granger Causality Test	The existence of the EKC hypothesis and bidirectional causality between FT and CO2 emissions were detected.
Sarkodie and Strezov [30]	1986-2016	China, India, Iran, Indonesia and South Africa	Energy Consumption, Greenhouse gas emissions	Panel data regression with Driscoll-Kraay standard errors, and U test estimation approach	It was concluded that energy consumption had a strong positive effect on greenhouse gas emissions.

METHODOLOGY

This article aims to examine the relationship between ecological footprint and energy consumption using a panel of 10 newly industrialized countries. With globalization, these countries are likely to increase their economic and financial relations with each other. Ignoring cross-sectional dependence across units may lead to misleading empirical results. In order to avoid this problem, cross-sectional dependence between units was tested.

Empirical Model and Dataset

The effect of energy consumption on ecological footprint was examined for 10 newly industrialized countries using annual data covering the period 1970-2022: Brazil, China, India, Indonesia, Malaysia, Mexico, the Philippines, South Africa, Thailand, and Türkiye. The panel version of the empirical model employed can be written as follows:

$$\ln EF_{it} = \varphi_0 + \varphi_1 \ln EC_{it} + \mu_{it} \quad (1)$$

where t , i and μ_{it} denote the time dimension, cross-sectional dimension, and residual terms, respectively. In addition, $\ln EF$ signifies the logarithm of the ecological footprint per capita, and $\ln EC$ signifies the logarithm of energy consumption per capita. Ecological footprint data was obtained from Global Footprint Network [36] and energy consumption data was obtained from the British Petroleum Statistical Review of World Energy [37].

Cross-Section Dependence Test

Breusch and Pagan's [34] LM (Lagrange Multiplier) test was applied to test the cross-sectional dependence between the units forming the panel. With the Breusch and Pagan [34] LM test, valid results are obtained when N is small and T is large. For the LM test, the equation can be written as follows:

$$y_{it} = \alpha_i + \beta_i x_{it} + \mu_{it} \quad (2)$$

$$i = 1, 2, \dots, N; \quad t = 1, 2, \dots, T \quad (3)$$

where, i refers to the cross-sectional dimension, and t refers to the time series dimension. If we define x and y variables within the scope of this study, x indicates energy consumption, and y indicates ecological footprint. The null hypothesis of the LM test is as shown below:

$$H_0: Cov(\mu_{it}, \mu_{jt}) = 0 \quad \text{for all } t \text{ and } i \neq j \quad (4)$$

and the alternative hypothesis is

$$H_a: Cov(\mu_{it}, \mu_{jt}) \neq 0 \quad \text{for at least one } i \neq j \quad (5)$$

LM test statistic to test the null hypothesis stating that there is no cross-sectional dependence:

$$LM = T \sum_{i=1}^{N-1} \sum_{j=i+1}^N \hat{\rho}_{ij} \quad (6)$$

The modified LM test formulation is as follows:

$$LM_{adj} = \sqrt{\left(\frac{2T}{N(N-1)}\right)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \hat{\rho}_{ij} \frac{(T-k)\hat{\rho}_{ij}^2 - \mu_{Tij}}{\sqrt{v_{Tij}^2}} \quad (7)$$

Panel Unit Root Test

Pesaran [4] developed a new panel unit root test by taking the difference from the cross-sectional averages of standard ADF (Augmented Dickey-Fuller) regressions. The linear equation of the Pesaran [4] CIPS (Cross-sectionally augmented Im-Pesaran-Shin) test used in the presence of cross-sectional dependence is as follows:

$$\Delta x_{it} = \alpha_i + \rho_i x_{i,t-1} + c_i \bar{x}_{t-1} + d_i \Delta \bar{x} + \varepsilon_{it} \quad (8)$$

where $\bar{x}_{t-1} = (1/N) \sum_{i=1}^N x_{i,t-1}$ and $\Delta \bar{x}_t = (1/N) \sum_{i=1}^N \Delta x_{it}$, ε_{it} are regression errors.

The cross-sectional average of the lagged values, $\bar{x}_{t-1}$, and the first differences of the individual series, $\Delta \bar{x}_t$, are included in the model as unobservable common factors. In the unit root test, which he based on the individual CADF tests referred to as CIPS, Pesaran [4] derived a generalized version of the t-bar test proposed by IPS, and its formulation is as follows:

$$CIPS = \frac{1}{N} \sum_{i=1}^N CADF_i \quad (9)$$

where, $CADF_i$ is the cross-sectionally augmented Dickey-Fuller test statistic for the i th cross-section unit given by the t-ratio of ρ_i in the CADF regression.

Panel Cointegration Test

In this study, the error correction-based panel cointegration test developed by Westerlund [5] was employed. In this test, the error correction model, based on the assumption that all variables are integrated of order one, is as follows:

$$\Delta z_{it} = \delta'_i d_t + \theta_i (z_{i(t-1)} - \beta'_i y_{i(t-1)}) + \sum_{j=1}^m \theta_{ij} \Delta z_{i(t-j)} + \sum_{j=0}^m \varphi_{ij} \Delta y_{i(t-j)} + \omega_{it} \quad (10)$$

where $d_t = (1-t)'$ contains the deterministic components, and $\delta'_i = (\delta_{1i}, \delta_{2i})'$ includes the associated vector of parameters.

In order to estimate the error correction parameter θ_i by using the least squares method, the above equation can be rewritten as follows:

$$\Delta z_{it} = \delta'_i d_i + \theta_i (z_{i(t-1)} - \pi'_i y_{i(t-1)}) + \sum_{j=1}^m \theta_{ij} \Delta z_{i(t-j)} + \sum_{j=0}^m \varphi_{ij} \Delta y_{i(t-j)} + \omega_{it} \quad (11)$$

Westerlund [5] developed four test statistics for the least squares estimation of θ_i . Two of these are called group mean statistics and are expressed as follows:

$$G_\tau = \frac{1}{N} \sum_{i=1}^N \frac{\theta_i}{SE(\hat{\theta}_i)} \quad (12)$$

$$G_\alpha = \frac{1}{N} \sum_{i=1}^N \frac{T\theta_i}{\theta_{i(1)}} \quad (13)$$

The null hypothesis of the G_τ and G_α test statistics states that there is no cointegration relationship, while the alternative hypothesis states that there is a cointegration relationship for at least one of the units forming the panel. The other two test statistics are called the panel statistics and are expressed as follows:

$$P_\tau = \frac{\theta_i}{SE(\hat{\theta}_i)} \quad (14)$$

$$P_\alpha = T\hat{\theta} \quad (15)$$

The null hypothesis of the P_τ and P_α test statistics states that there is no cointegration relationship, while the alternative hypothesis states that there is a cointegration relationship for all the units comprising the panel.

Panel Causality Tests

Dumitrescu and Hurlin [8] developed the Granger [35] causality test for heterogeneous panels. This causality test addresses two dimensions of heterogeneity: (1) the heterogeneous regression model to be used for testing Granger causality and (2) the heterogeneous causality relationships. The linear equation of Dumitrescu and Hurlin's [11] causality test is as follows:

$$\Delta z_{it} = \alpha_i + \sum_{m=1}^M \gamma_i^{(m)} z_{i,t-m} + \sum_{m=1}^M \beta_i^{(m)} \gamma_{i(t-j)} + \omega_{it} \quad (16)$$

where it indicates that the y and z series are stationary for T periods and N units. The autoregressive parameters $\gamma_i^{(m)}$ and regression coefficient estimates $\beta_i^{(m)}$ are assumed to vary by units. The null hypothesis states that there is no causal relationship between the series for any units under homogeneous panel causality and is shown as follows:

$$H_0: \beta_i = 0 \quad \forall i = 1, 2, \dots, N \quad (17)$$

$$H_0 \neq \beta_i \neq 0 \quad \forall i = 1, 2, \dots, N \quad (18)$$

The alternative hypothesis indicates that the model is heterogeneous; β_i s are evaluated across units. The alternative hypothesis, which states that β_i s are different from zero and there is a causality relationship between some units, is formulated as follows:

$$H_a: \beta_i = 0 \quad \forall i = 1, 2, \dots, N_1 \quad (19)$$

$$\beta_i \neq 0 \quad \forall i = N_1 + 1, \dots, N \quad (20)$$

As a result, under the alternative hypothesis, with $N_1 < N$, there is no causality relationship from the independent variable to the dependent variable. Here, N_1 is unknown, but the condition $0 \leq N_1/N < 1$ must be satisfied. If $N_1 = N$, the alternative hypothesis means that there is no causality for any of the units forming the panel, which is the same as the null hypothesis.

The Wald statistics used in the causality test for each cross-sectional dimension to test the null hypothesis is as follows:

$$W_{N,T}^{HNC} = \frac{1}{N} \sum_{i=1}^N W_{i,T} \quad (21)$$

where, $W_{i,T} (W_{i,T} = \hat{\theta}'_i R' [\sigma_i^2 R (Z'_i Z_i)^{-1} R'] R \hat{\theta}_i)$ represents the Wald test statistics for each unit.

When T and N go to infinity for the $W_{N,T}^{HNC}$ statistic, the test statistic can be written as follows:

$$Z_{N,T}^{HNC} = \sqrt{\frac{N}{2M}} (W_{N,T}^{HNC} - M) \rightarrow N(0,1) \quad (22)$$

If N is large,

$$Z_{N,T}^{HNC} = \sqrt{\frac{N}{2 \times K} \times \frac{(T-2K-5)}{(T-K-3)}} \times \left[\frac{(T-2K-3)}{(T-2K-1)} W_{N,T}^{HNC} - K \right] \rightarrow N(0,1) \quad (23)$$

$$\beta_i \neq 0 \quad \forall i = N_1 + 1, \dots, N \quad (24)$$

statistics can be used.

EMPIRICAL FINDINGS

The cross-sectional dependence and heterogeneity of slope coefficients affect the estimates between ecological footprint (EF) and energy consumption (EC). Therefore, it is essential to test for cross-sectional dependence and homogeneity before the analysis. Table 2 shows the results of cross-sectional dependence and homogeneity tests. These tests are the LM test introduced by Breusch & Pagan [34] and the CD Test by Pesaran, Ullah, & Yamagata [38] for cross-sectional dependence; and the Delta Test by Pesaran & Yamagata [39] for homogeneity analysis. According to the results in Table 2, all test statistics are statistically significant at the 1% level, indicating the presence of both cross-sectional dependence and heterogeneity across the panel.

Table 2. Cross-section dependence and homogeneity tests

Test	Statistics (p-value)
LM	231 (0.0000)*
LM _{adj}	79.73 (0.0000)*
LM _C	6.93 (0.0000)*
Δ	36.393 (0.0000)*
Δ_{adj}	37.468 (0.0000)*

* The values in parentheses indicate probabilities.

CIPS unit root test was applied to test the stationarity of ecological footprint and energy consumption, and the results are shown in Table 3. It was concluded that, in the presence of cross-section dependence, the ecological footprint and energy consumption variables contained unit roots in the constant and constant & trend versions. When Table 3 is examined, it can be seen that ecological footprint and energy consumption are stationary in the first differences. Once it was understood that both variables were integrated at the I(1) level, the error correction-based panel cointegration test developed by Westerlund [5] was applied to examine whether there was a long-run relationship between ecological footprint and energy consumption.

Table 3. Pesaran (2007) Panel Unit Root Test

	Constant	Constant and Trend
Level		
ln EF	-2.040	-2.367
ln EC	-1.994	-2.429
1st Difference		
Δ ln EF	-5.050*	-5.104*
Δ ln EC	-3.805*	-3.770*

* indicates that the null hypothesis was rejected at the 1% significance level. ln EF represents the natural logarithm of ecological footprint, while ln EC represents the natural logarithm of energy consumption.

Table 4 shows the panel cointegration test results. When the results are analyzed, the null hypothesis indicating that there is no cointegration between the variables is rejected according to the group statistics (G_t and G_a are at 5% and 10% significance levels, respectively) and panel statistics (P_t and P_a %1 are at 1% significance level). This supports the hypothesis that energy consumption had an impact on the ecological footprint over the period 1970-2022 for newly industrialized countries.

The existence of a panel cointegration relationship between ecological footprint and energy consumption in 10 newly industrialized countries allows us to examine the panel effects of energy consumption on the ecological footprint. Table 5 shows the results of Eberhardt and Teal's [6] AMG estimator and Pesaran's [7] CCE estimator, which are utilized

to estimate the long-run cointegration relationship for heterogeneous panels. When the AMG estimator results in the table are examined, energy consumption positively affects the ecological footprint for Brazil (at 1% significance level), China (at 1% significance level), India (at 1% significance level), Indonesia (at 1% significance level), Malaysia (at 1% significance level), Mexico (at 1% significance level), the Philippines (at 1% significance level), South Africa (at 1% significance level), Thailand (at 1% significance level), and Türkiye (at 1% significance level). Furthermore, the AMG estimator panel statistic also shows that energy consumption has a positive and significant (at the 1% significance level) effect on the ecological footprint. This situation shows that the increase in energy consumption strongly increases the ecological footprint.

Table 4. Westerlund (2007) Cointegration Test

	Statistic	Z-statistic	Probability	Robust Probability
G_t	-2.579	-2.797	0.003	0.030
G_a	-11.017	-2.226	0.013	0.080
P_t	-10.533	-5.795	0.000	0.000
P_a	-13.771	-6.489	0.000	0.000

Optimal lag/lead length determined by the Akaike information criterion, with a maximum lag/lead length of 1. The values robust to cross-sectional dependence were obtained from a distribution with 100 replications.

Table 5. AMG and CCE estimation results

Countries	AMG		CCE	
	Coefficient	Z	Coefficient	Z
Brazil	0.103	2.75*	0.696	7.59*
China	0.669	93.42*	0.677	26.60*
India	0.454	44.00*	0.633	16.11*
Indonesia	0.210	10.30*	-0.372	-4.80*
Malaysia	0.407	18.05*	0.641	5.24*
Mexico	0.773	9.63*	1.021	7.53*
The Philippines	0.296	6.50*	0.409	5.16*
South Africa	0.413	3.19*	0.316	2.37**
Thailand	0.461	31.72*	0.744	10.15*
Türkiye	0.423	24.15*	0.468	4.45*
Panel	0.421	6.75*	0.523	4.46*

* and ** indicate 1% and 5% significance levels, respectively. AMG: Eberhardt and Teal (2010) Augmented Mean Group Estimator; CCE: Pesaran (2006) Common Correlated Effects Mean Group Estimator

Table 6. Dumitrescu and Hurlin (2012) Granger Causality Test

Null Hypothesis	EC is not the cause of EF.	EF is not the cause of EC.
	EC≠EF	EF≠EC
Panel Test Statistics		
W^{HNC}	6.9014*	1.6587
W_{NT}^{HNC}	13.1958*	1.4730
W_N^{HNC}	12.1669*	1.2796

* denotes at 1% significance level.

When the results of the CCE estimator are examined, energy consumption positively affects the ecological footprint in Brazil (at 1% significance level), China (at 1% significance level), India (at 1% significance level), Malaysia (at 1% significance level), Mexico (at 1% significance level), the Philippines (at 1% significance level), South Africa (at 5% significance level), Thailand (at 1% significance level), and Türkiye (at 1% significance level). Although there is a significant relationship for Indonesia (at the 1% significance level), its coefficient is negative. Moreover, the CCE estimator panel statistic also shows that energy consumption has a positive and significant (at the 1% significance level) effect on the ecological footprint. This reveals that the increase in energy consumption strongly increases the ecological footprint.

In order to examine the causality relationship between ecological footprint and energy consumption, Dumitrescu and Hurlin's [8] Granger Causality Test was conducted, and the results are presented in Table 6. Looking at the results, for the null hypothesis that EC is not the cause of EF, the W^{HNC} , W_{NT}^{HNC} and W_N^{HNC} panel test statistics are significant (at a 1% significance level). However, for the null hypothesis that EF is not the cause of EC, the W^{HNC} , W_{NT}^{HNC} and W_N^{HNC} panel test statistics are not significant. Consequently, there is a unidirectional causal relationship running from energy consumption to ecological footprint.

CONCLUSIONS AND POLICY IMPLICATIONS

Recently, with the increasing impacts of growing energy consumption on the environment due to globalization in newly industrializing countries such as India, it is observed that there are studies examining energy consumption and its impact on the environment [32]; [40]. There are very few studies that examine the negative effects of energy consumption on the environment by addressing the impact of energy consumption on the ecological footprint, and to the best of our knowledge, no study has systematically addressed the causality between ecological footprint and energy consumption for newly industrializing countries. Based on this research gap, the relationship between ecological footprint and energy consumption was analyzed for newly industrialized countries. In this context, the study fills a research gap in the existing literature by analyzing the negative impacts on the environment due to the increase in energy consumption within the scope of ecological footprint.

The dataset of this study, which examines the relationship between ecological footprint and energy consumption, covers the period 1970-2022 for 10 newly industrialized countries. The countries that make up the panel are Brazil, China, India, Indonesia, Malaysia, Mexico, the Philippines, South Africa, Thailand, and Türkiye. The stationarity of the variables was tested utilizing Pesaran's [4] CIPS test, which can be used in the presence of cross-sectional dependence. The existence of a long-run relationship between ecological footprint and energy consumption was determined by applying the Westerlund [5] cointegration test. The long-run cointegration coefficient estimates for heterogeneous panels were calculated using the Pesaran [7] CCE estimator and the Eberhardt and Teal [6] AMG estimator. The causality relationships between the variables were analyzed by applying Dumitrescu and Hurlin's [11] causality test.

The empirical results prove the existence of a relationship between ecological footprint and energy consumption in 10 newly industrialized countries. According to the results of Eberhardt and Teal's [12] AMG estimator, energy

consumption has a positive and significant effect on the ecological footprint of Brazil, China, India, Indonesia, Malaysia, Mexico, the Philippines, South Africa, Thailand, and Türkiye. According to the results of Pesaran's [24] CCE estimator, energy consumption positively affects the ecological footprint of Brazil, China, India, Malaysia, Mexico, the Philippines, South Africa, Thailand, and Türkiye. Dumitrescu and Hurlin's [8] causality test results reveal unidirectional causality from energy consumption to ecological footprint.

In line with the findings of this study, important policy recommendations can be made. The increase in energy consumption has a negative impact on the environment in almost all countries in the dataset. These results show that the energy sources, the use of which is increasing with globalization, are non-renewable energy sources. From a policy perspective, it can be said that the increasing demand for energy resources should be met by renewable energy resources. To achieve this, governments should implement comprehensive incentive schemes to promote the adoption of renewable energy technologies, including subsidies for green investments, tax reductions for clean energy producers, and grants for R&D in environment-related technologies. In addition, regulatory frameworks that prioritize energy efficiency standards and the integration of renewables into national energy mixes should be strengthened. Otherwise, the world will face longer-term environmental consequences in terms of climate change and global warming with the continued use of non-renewable energy resources.

This study is subject to several limitations. First, the analysis focuses on total energy consumption without differentiating between energy sources. Since the environmental effects of renewable and non-renewable energy sources differ significantly, future studies may yield more detailed insights by disaggregating energy types. Second, although the ecological footprint provides a broader indicator of environmental pressure compared to conventional proxies such as CO₂ emissions, it may not fully capture sector-specific environmental effects. Future research could address these limitations by incorporating more disaggregated environmental indicators and energy data.

DATA AVAILABILITY STATEMENT

The authors confirm that the data that supports the findings of this study are available within the article. Raw data that support the finding of this study are available from the corresponding author, upon reasonable request.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest with any individual, institution, or organization in the preparation, evaluation, or publication of this study.

USE OF AI FOR WRITING ASSISTANCE

Not declared.

ETHICS

There are no ethical issues with the publication of this manuscript.

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